

# 2.10.76

EE25BTECH11048 - Revanth Siva Kumar.D

**Question** Find the equation of the line joining the points (3, 1) and (9, 3).

**Solution :**

Given

$$\mathbf{A} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \mathbf{B} = \begin{pmatrix} 9 \\ 3 \end{pmatrix} \quad (1)$$

Let us assume line equation to be:

$$\mathbf{n}^T \mathbf{x} = c \quad (2)$$

We get the line equation on solving

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix}^T \mathbf{n} = c \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The line passes through the points from (1) substituting, we get:

$$\begin{pmatrix} 3 & 9 \\ 1 & 3 \end{pmatrix}^T \mathbf{n} = c \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (3)$$

$$\begin{pmatrix} 3 & 1 \\ 9 & 3 \end{pmatrix} \mathbf{n} = c \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (4)$$

Now by Gaussian Elimination solve:

$$\left( \begin{array}{cc|c} 3 & 1 & 1 \\ 9 & 3 & 1 \end{array} \right) \quad (5)$$

$$\begin{aligned} R_1 &\leftarrow \frac{1}{3}R_1 \\ \Rightarrow \left( \begin{array}{cc|c} 1 & \frac{1}{3} & \frac{1}{3} \\ 9 & 3 & 1 \end{array} \right) \end{aligned} \quad (6)$$

$$\begin{aligned} R_2 &\leftarrow R_2 - 9R_1 \\ \Rightarrow \left( \begin{array}{cc|c} 1 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & -2 \end{array} \right) \end{aligned} \quad (7)$$

By the assumption that line equation is  $\mathbf{n}^T \mathbf{x} = 1$  which doesn't pass through origin we are not getting any solution. So our assumption is wrong and origin lies on the line. So consider

$$\mathbf{n}^T \mathbf{x} = 0 \quad (8)$$

$c = 0$  because origin lies on the line and solving: so now, Assume the line equation:

$$\mathbf{n}^T \mathbf{x} = 0, \quad \mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix}$$

Line passes through points **A** and **B**

$$\mathbf{n}^T \mathbf{A} = 0 \implies 3n_1 + 1n_2 = 0 \quad (9)$$

$$\mathbf{n}^T \mathbf{B} = 0 \implies 9n_1 + 3n_2 = 0 \quad (10)$$

Matrix form:

$$\begin{pmatrix} 3 & 1 \\ 9 & 3 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (11)$$

Augmented matrix:

$$\left( \begin{array}{cc|c} 3 & 1 & 0 \\ 9 & 3 & 0 \end{array} \right) \quad (12)$$

$$\begin{aligned} R_1 &\leftarrow \frac{1}{3}R_1 \\ &\Rightarrow \left( \begin{array}{cc|c} 1 & \frac{1}{3} & 0 \\ 9 & 3 & 0 \end{array} \right) \end{aligned} \quad (13)$$

$$\begin{aligned} R_2 &\leftarrow R_2 - 9R_1 \\ &\Rightarrow \left( \begin{array}{cc|c} 1 & \frac{1}{3} & 0 \\ 0 & 0 & 0 \end{array} \right) \end{aligned} \quad (14)$$

From first row:

$$n_1 + \frac{1}{3}n_2 = 0 \implies n_1 = -\frac{1}{3}n_2 \quad (15)$$

$$(16)$$

Let,

$$n_2 = 3 \implies n_1 = -1 \quad (17)$$

$$\mathbf{n} = \begin{pmatrix} -1 \\ 3 \end{pmatrix} \quad (18)$$

$$\mathbf{n}^T \mathbf{x} = 0 \implies (-1 \quad 3)\mathbf{x} = 0 \quad (19)$$

The equation of the line passing through (3, 1) and (9, 3) is:

$$\boxed{(-1 \quad 3)\mathbf{x} = 0}$$

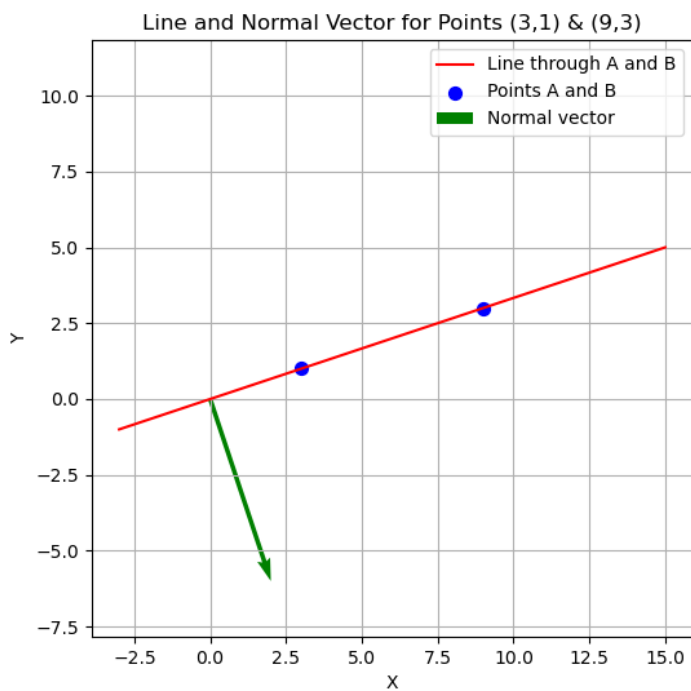


Fig. 1: PLOT